Quantum Annealing for Robust Financial Pattern Discovery

Outperforming conventional classic methods with Quantum L2 Principal Component Analysis in a $25+ Trillion Market

$|V\rangle$ x

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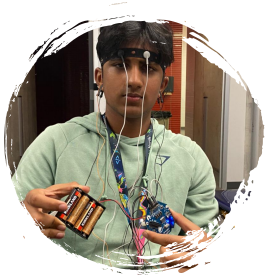
Electrical, Computer + Biomedical Engineering
Freshman

Santiago Lombardi
Electrical + Computer Engineering
Freshman

Rae Kapitonova
Mechanical Engineering
Freshman



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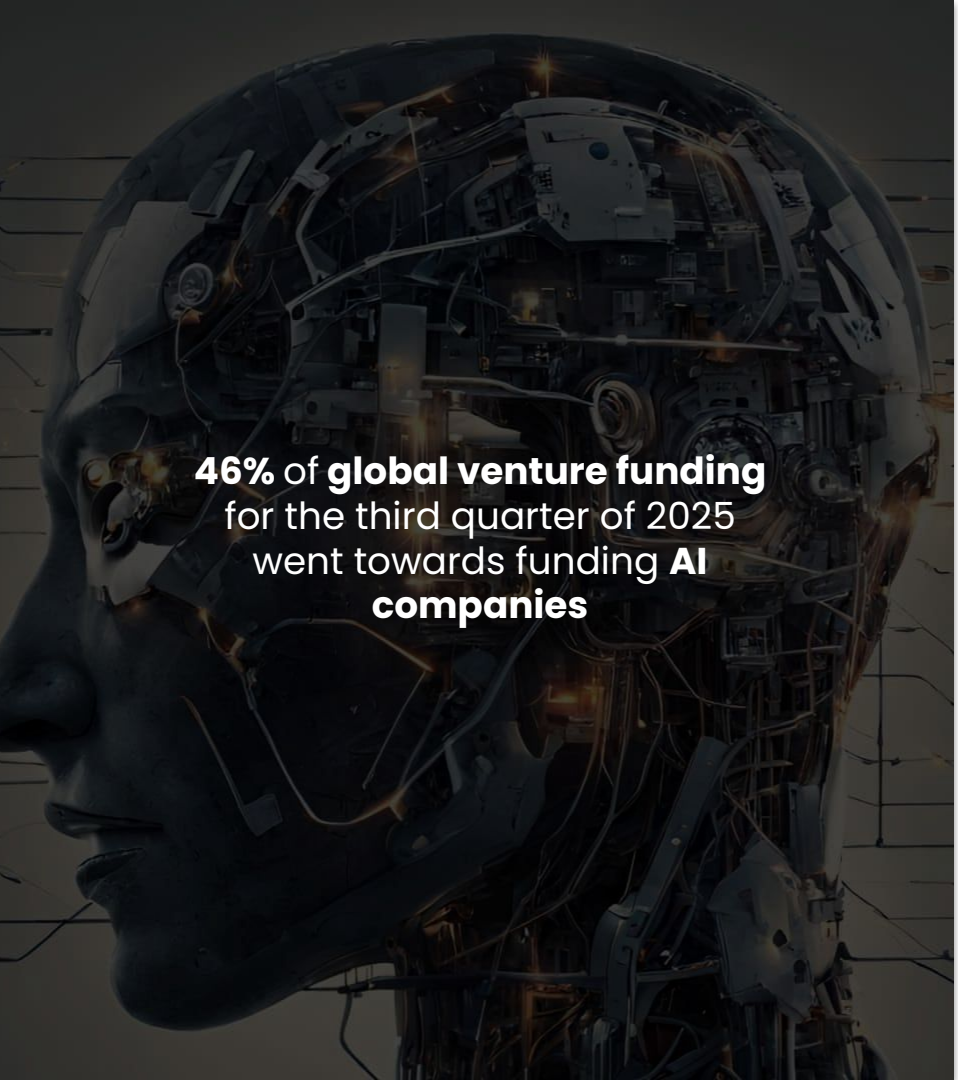
Freshman



46% of **global venture funding** for the third quarter of 2025 went towards funding **AI companies**

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U.S. fintech startups raised **$26 billion** in 2024, with **investment tech** being a top segment

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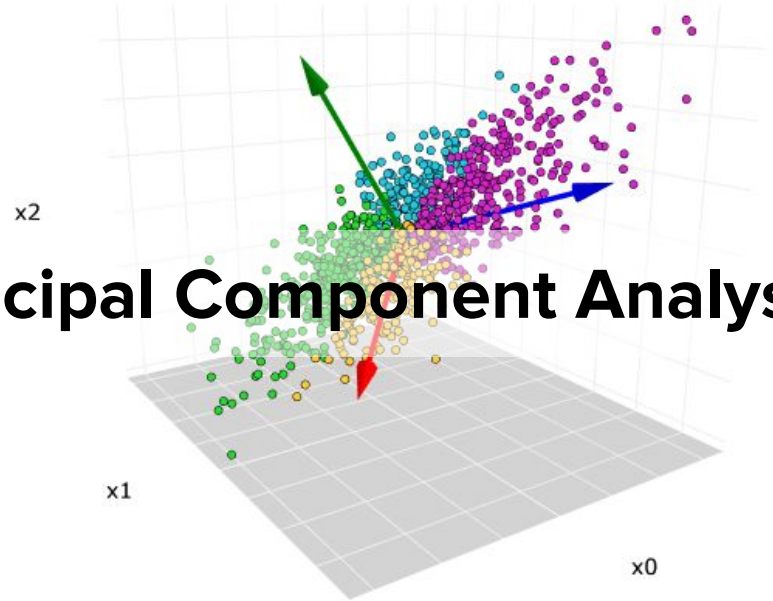
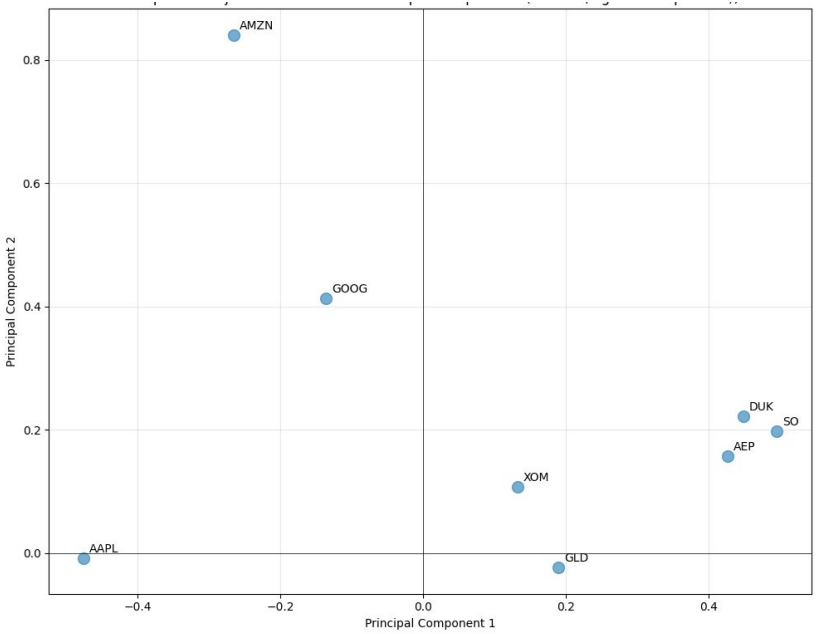
Collaboration = Perfect Growth Opportunity

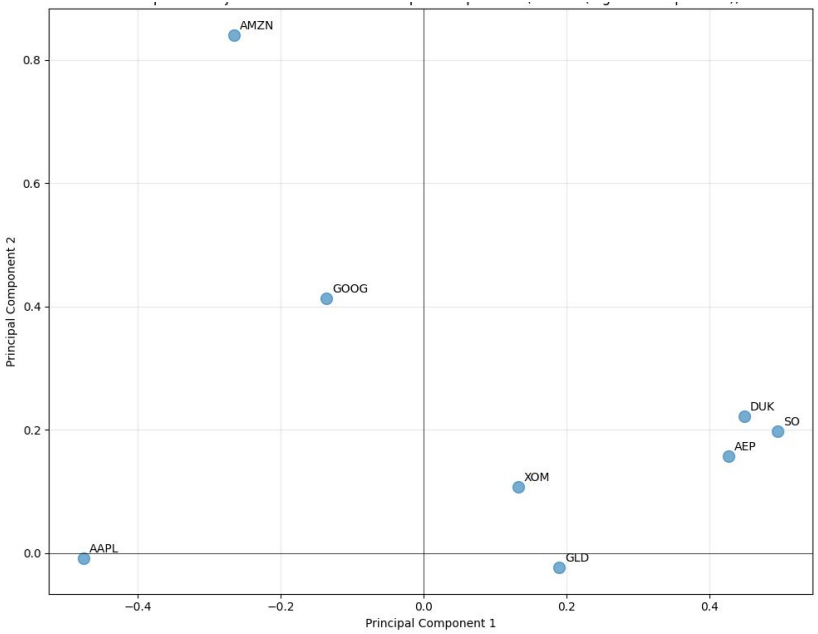
Principal Component Analysis

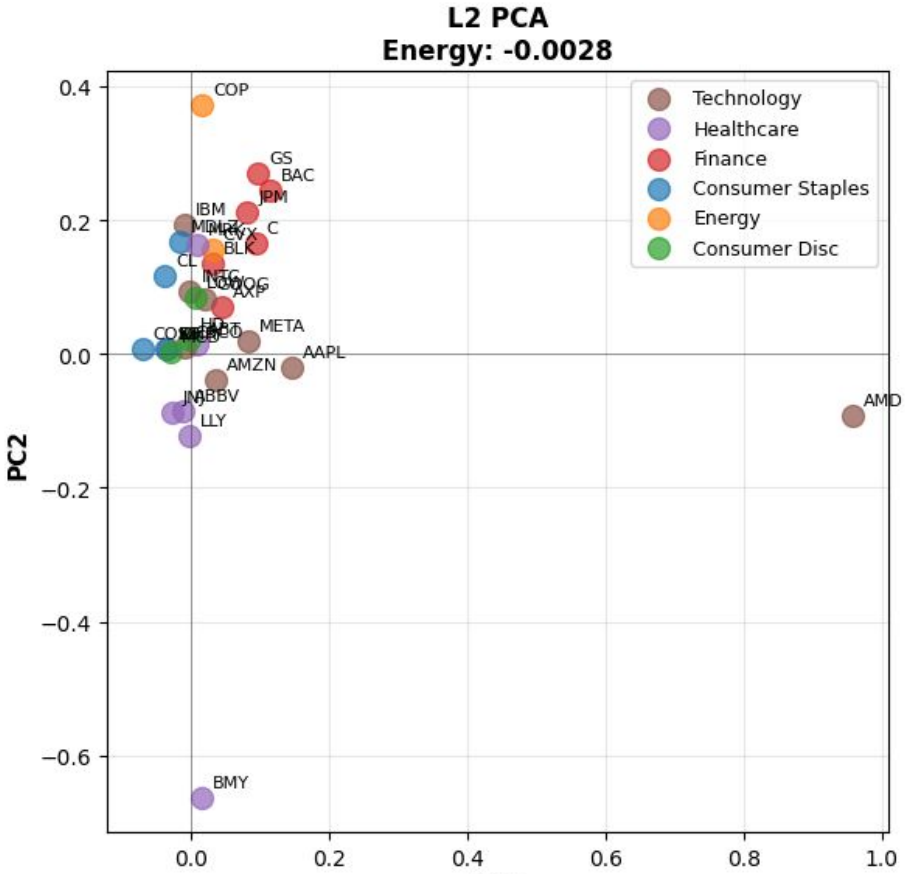


L2 Principal Component Analysis
→ Eigen-Decomposition

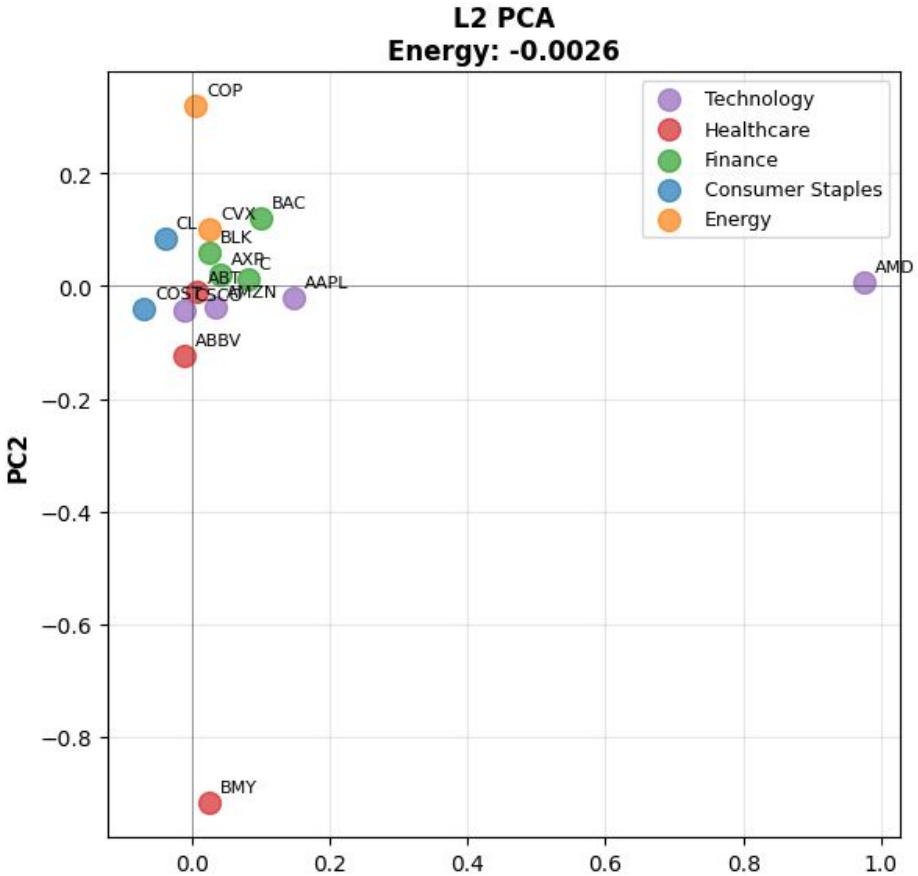


L2 Principal Component Analysis
→ Sklearn Machine Learning

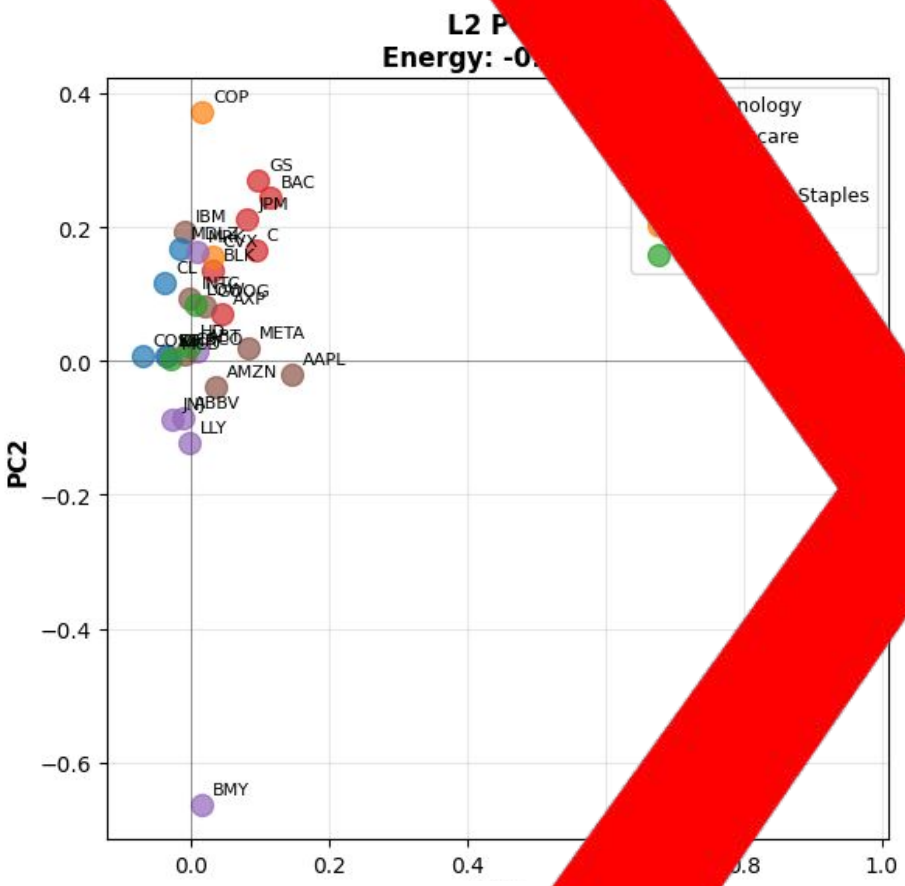




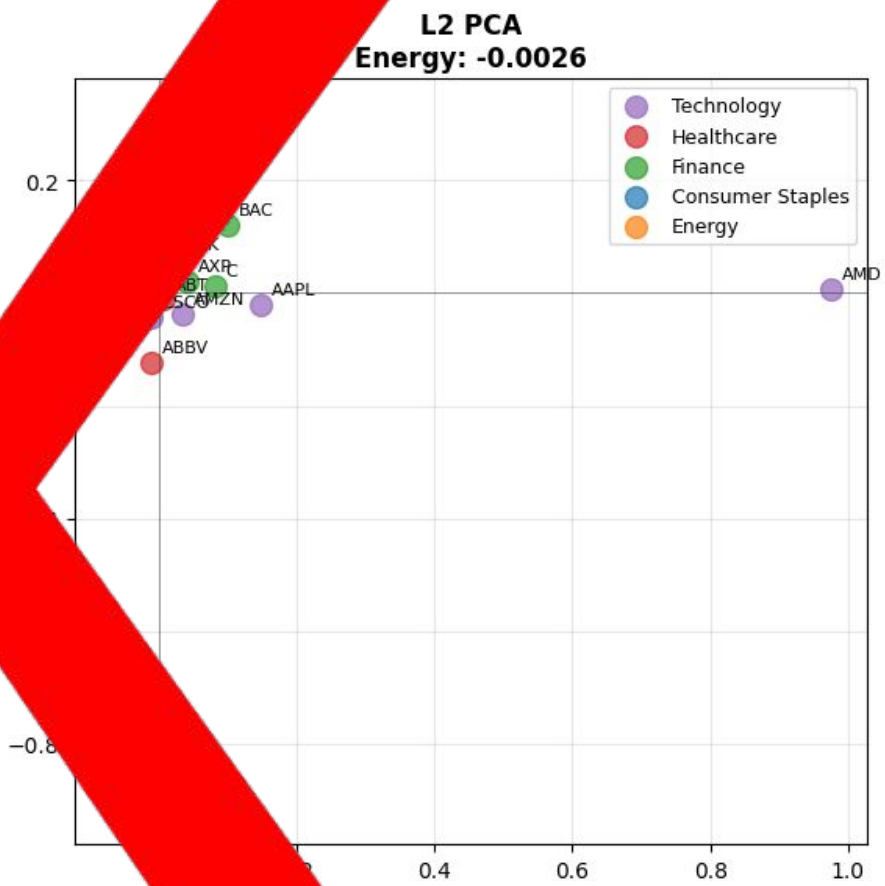
15 Tickers



30 Tickers



15 Tic



30 Tickers

L1 (Absolute Error) >>> L2 (Squared Error)

L2 = Sensitive to Extreme Outliers + Leads to Distorted Patterns

L1 = Robust to Outliers + Market Anomalies

Problem: L1 is NP-hard

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- Absolute value function

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- Absolute value function
- Many Local Minima

Problem: L1 is NP-hard

- Absolute value function
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- Not Differentiable

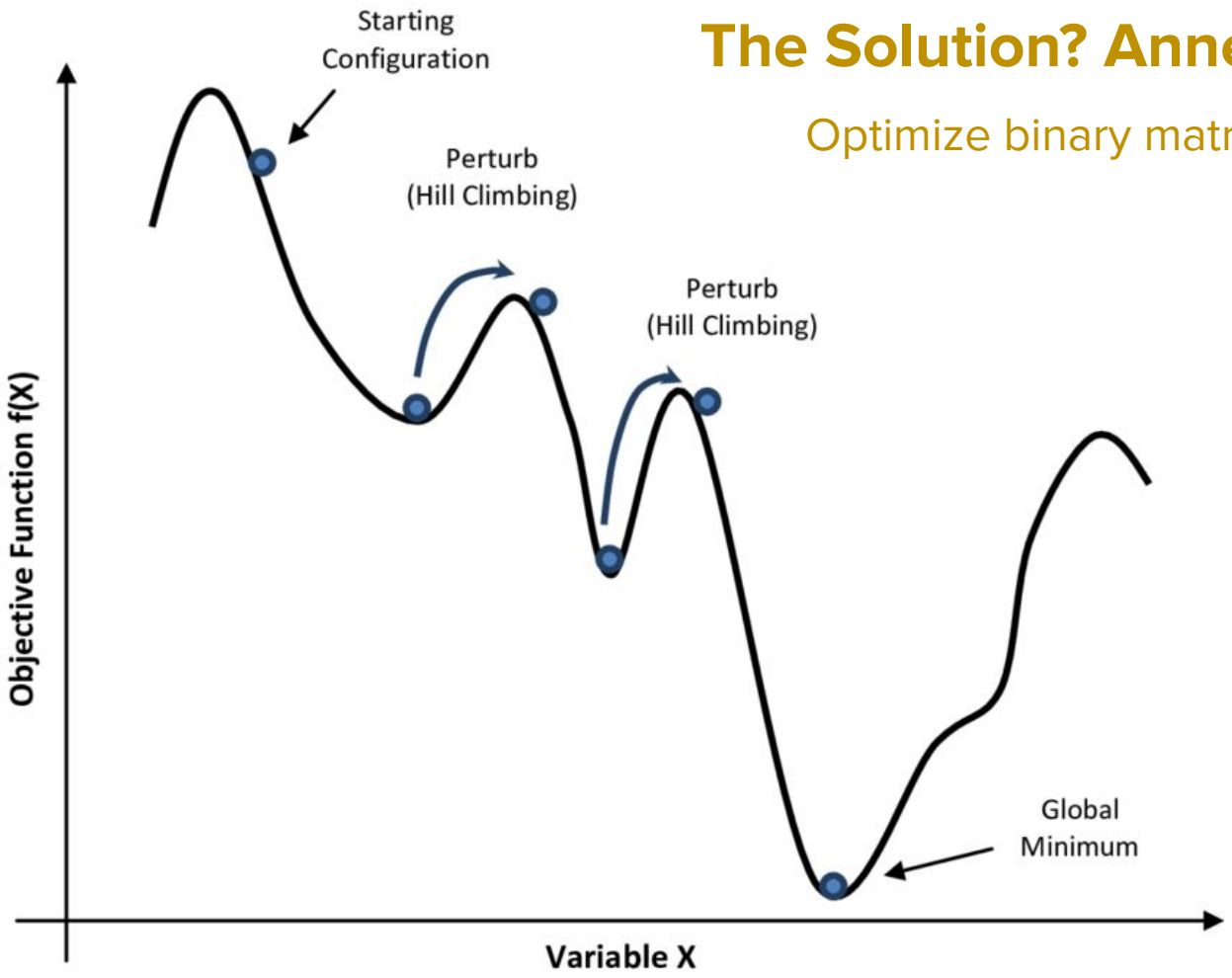
Problem: L1 is NP-hard

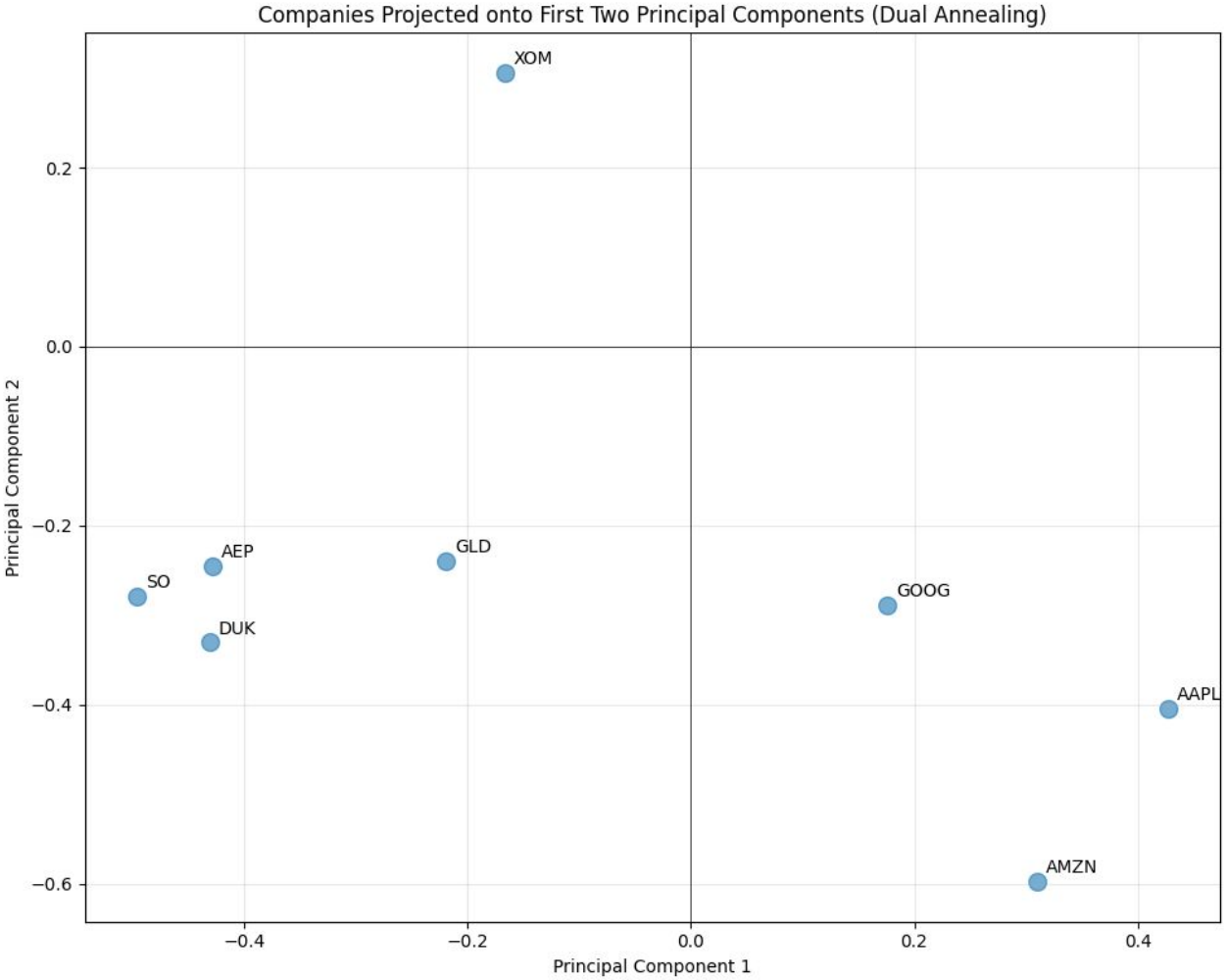
- Absolute value function
- Many Local Minima
- Not Differentiable

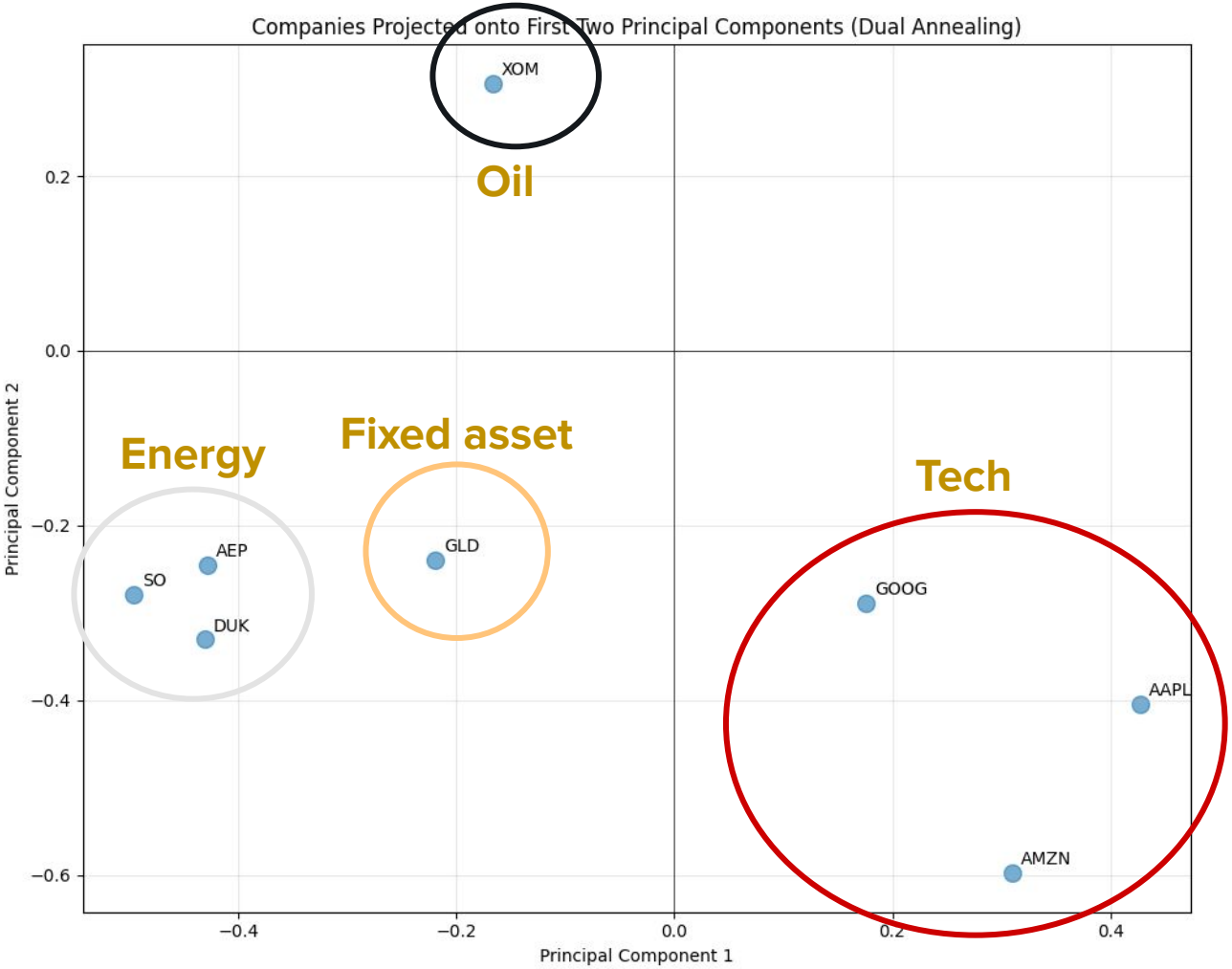
How can we solve this?

The Solution? Annealing

Optimize binary matrix




Companies Projected onto First Two Principal Components (Dual Annealing)
XOM
AEP
GLD
SO
DUK
GOOG
AAPL
AMZN
0.2
0.0
−0.2
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
Principal Component 2
Principal Component 1


Companies Projected onto First Two Principal Components (Dual Annealing)
XOM
Oil
Energy
Fixed asset
Tech
AEP
SO
DUK
GLD
GOOG
AAPL
AMZN
0.2
0.0
−0.2
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
Principal Component 2
Principal Component 1

But Libraries Aren’t Optimized for this Problem

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Continuous

vs.

Discrete optimization

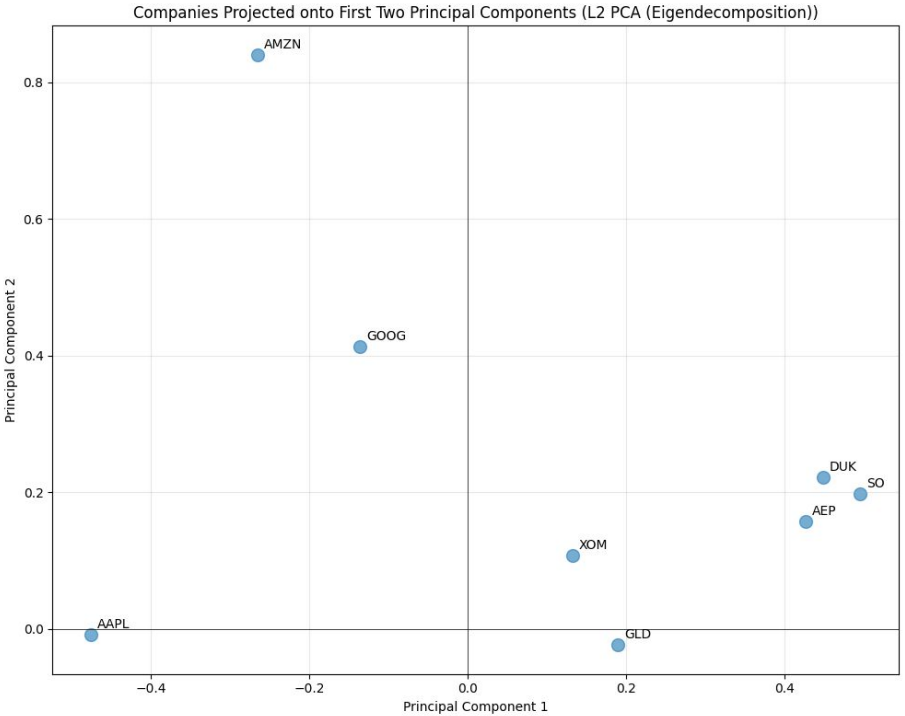
But Libraries Aren’t Optimized for this Problem

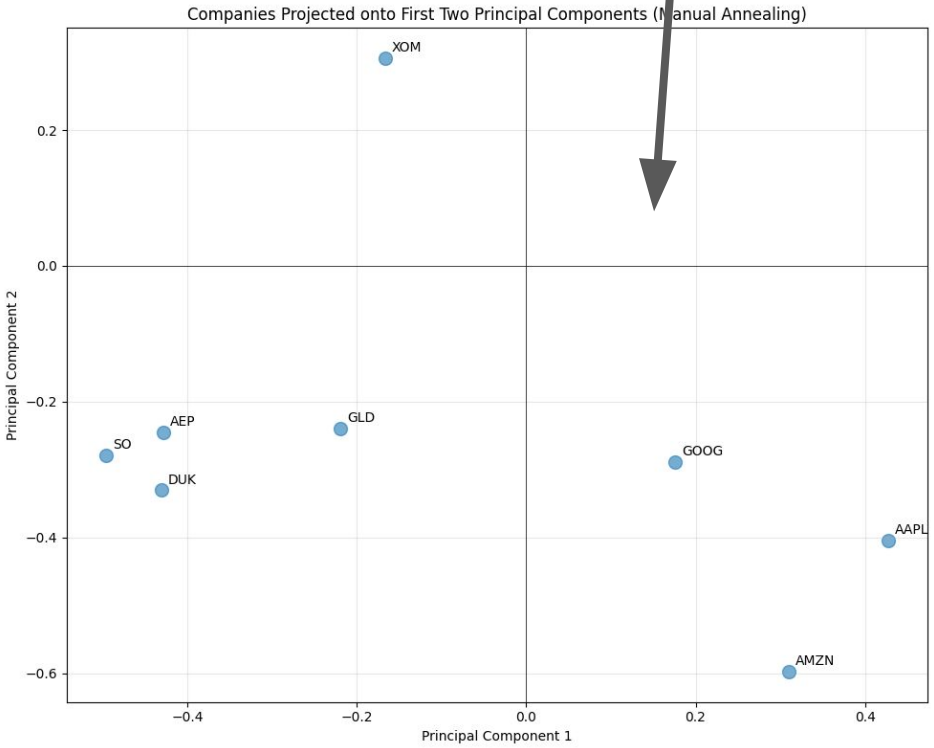
Continuous
vs.
Discrete
optimization

Allows for control
over:
Temperature function
Iteration steps

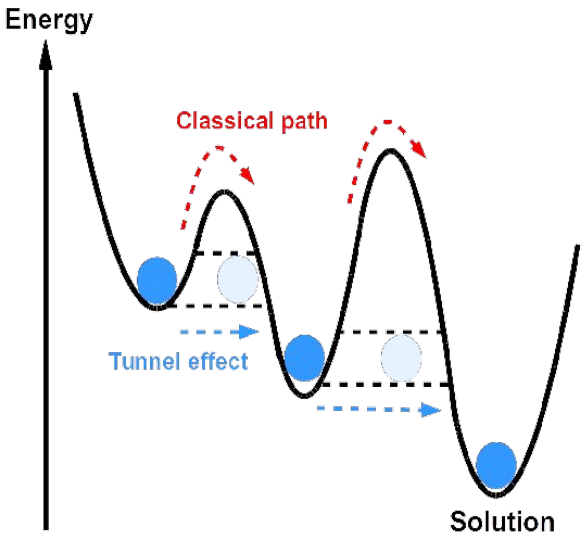
MANUAL ANNEALING!

Same graph as library annealing

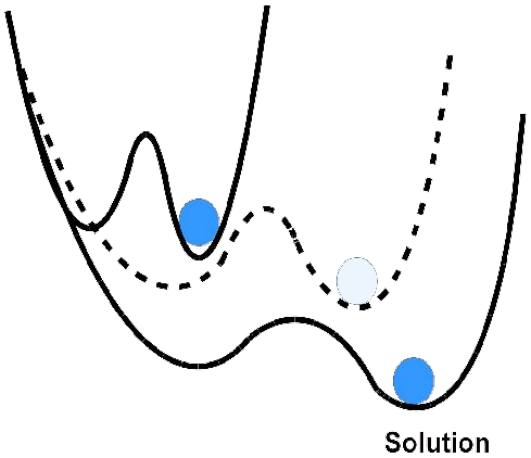




Why Quantum Annealing?



Quantum Tunnelling

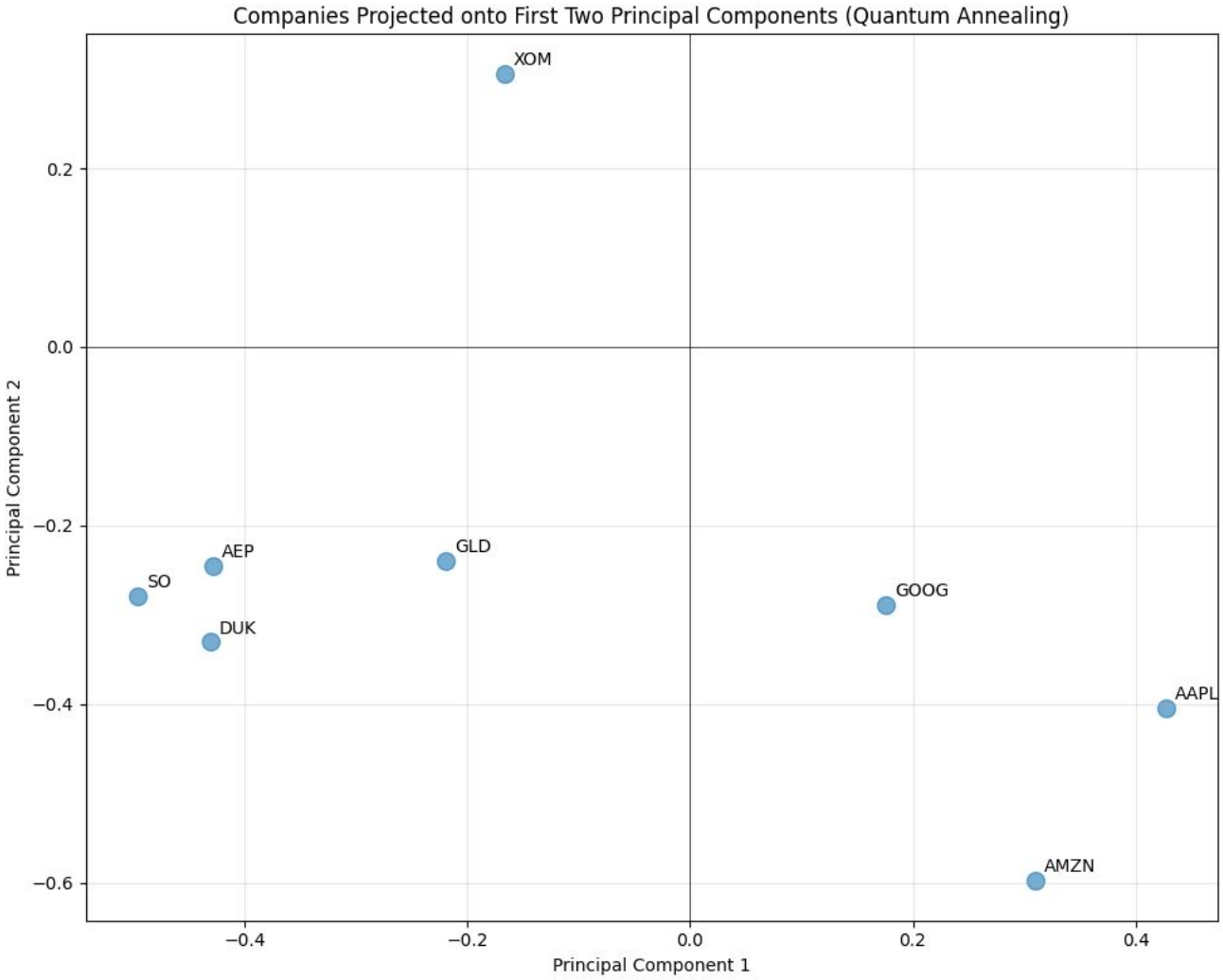


Adiabatic evolution

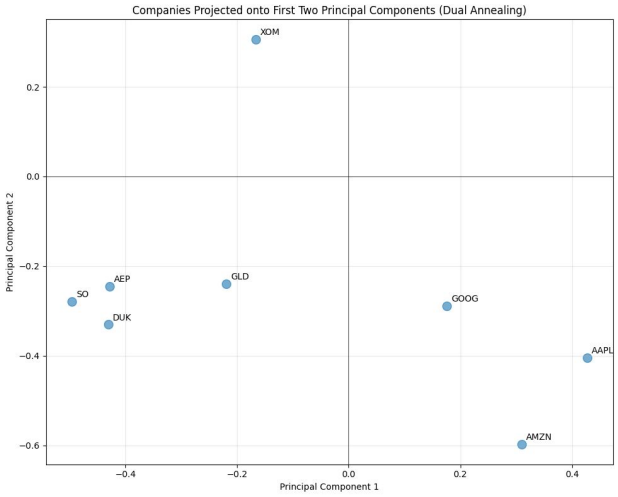
Why Quantum Annealing?

Frequent and Skinny Local Minima.

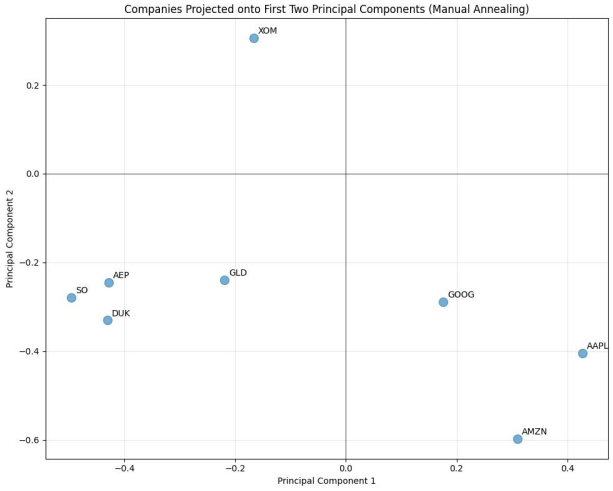
Implemented Using Ising Model.


Companies Projected onto First Two Principal Components (Quantum Annealing)
XOM
AEP
GLD
SO
DUK
GOOG
AAPL
AMZN
0.2
0.0
−0.2
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4
Principal Component 2
Principal Component 1

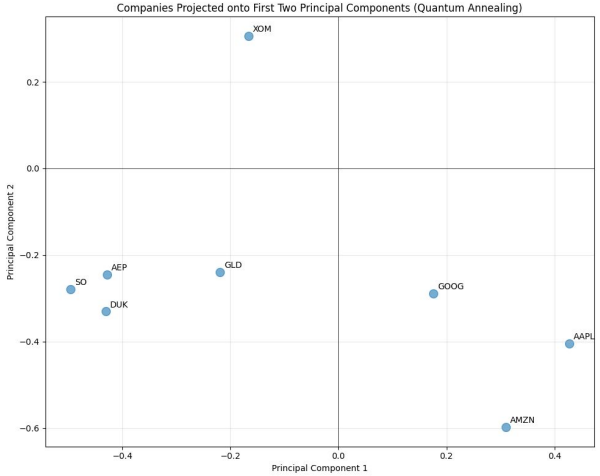
Same graphs!!!



Dual

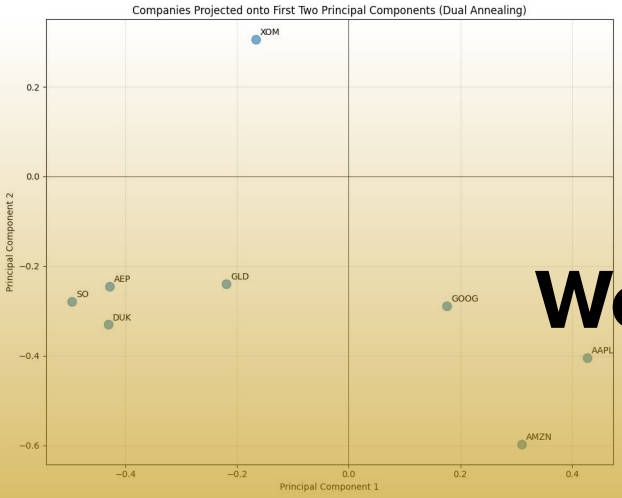


Manual

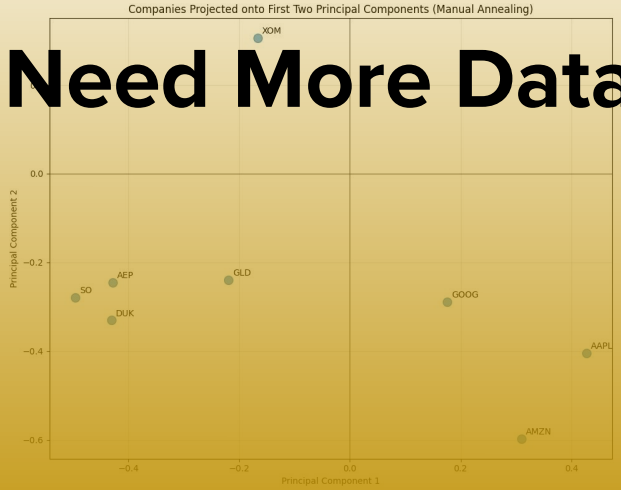


Quantum

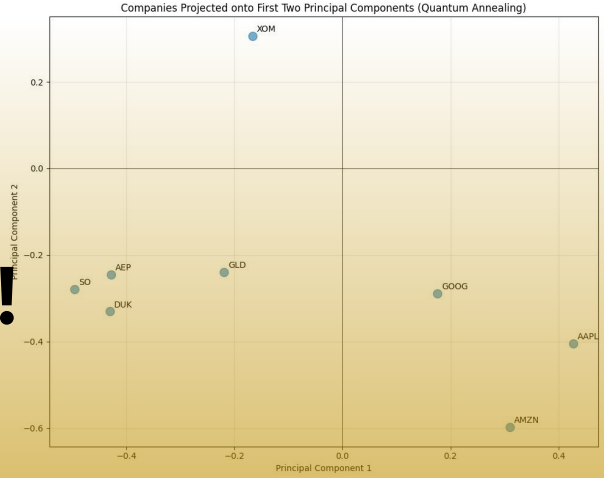
Same graphs!!!



Dual

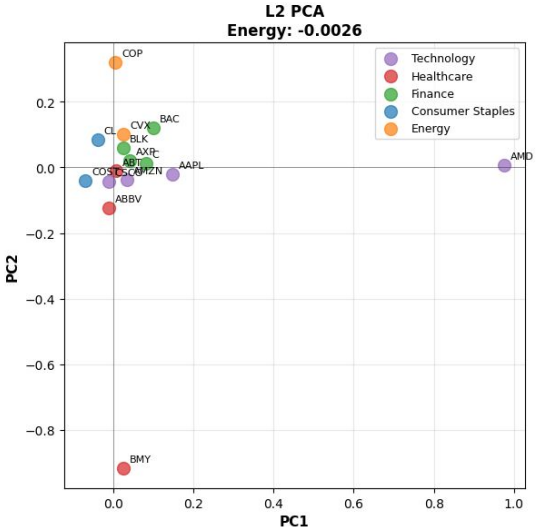


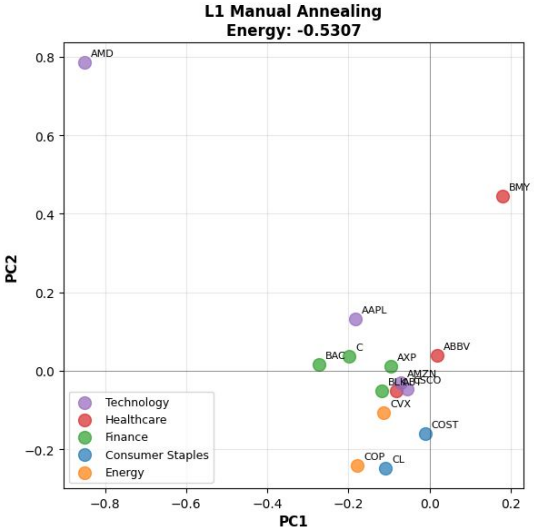
Manual

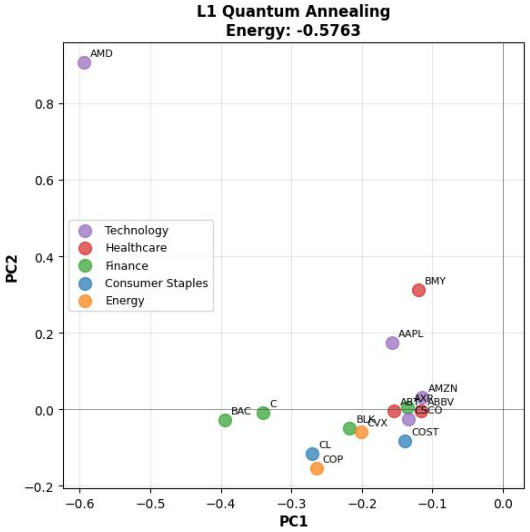


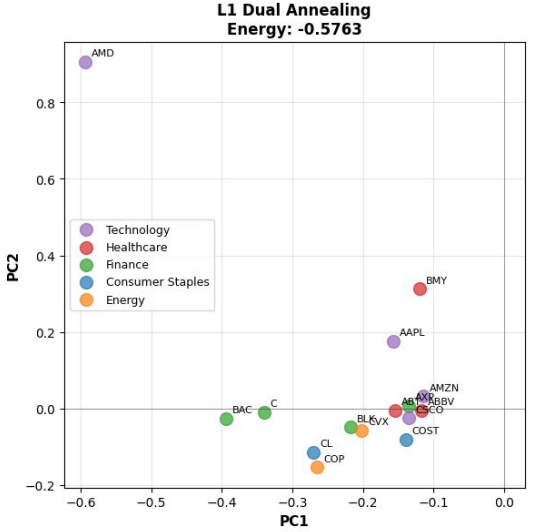
Quantum

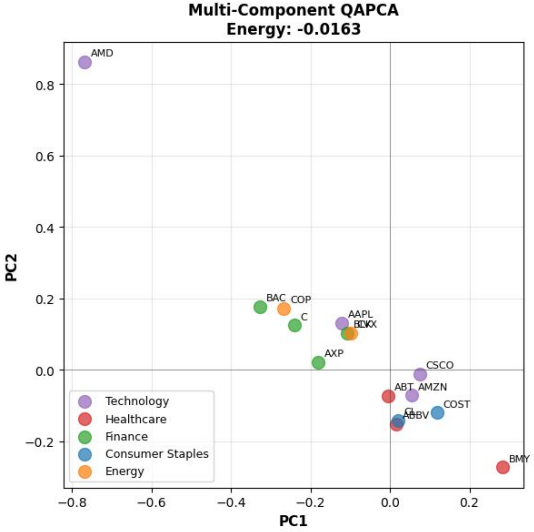
We Need More Data!


L2 PCA
Energy: -0.0026
Technology
Healthcare
Finance
Consumer Staples
Energy
COP
BAC
CVX
CL
BLK
AXP
C
AMD
AAPL
COST
CSCO
ABT
AMZN
ABBV
BMY
PC1
PC2

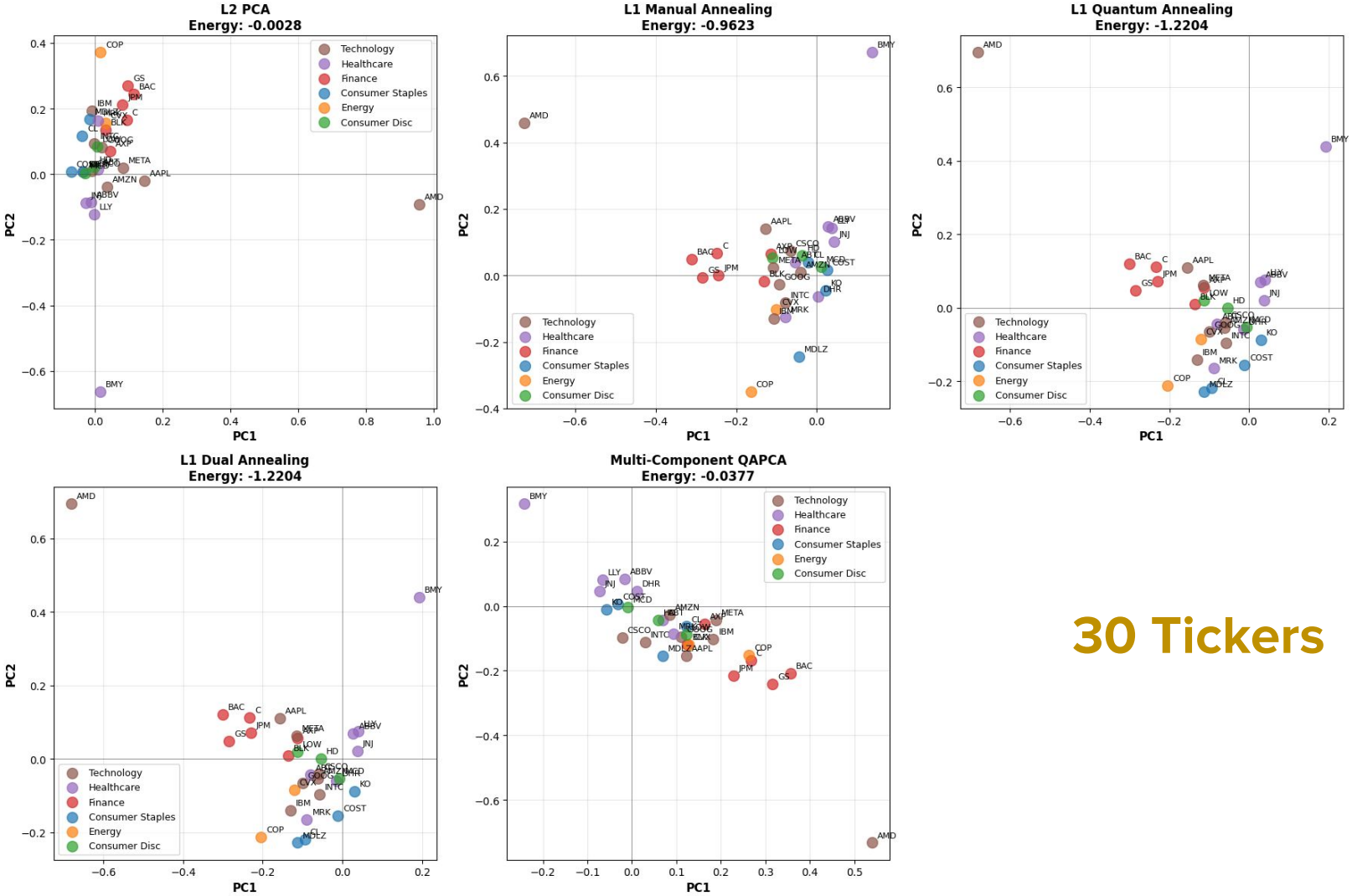

L1 Manual Annealing
Energy: -0.5307
AMD
BMY
AAPL
ABBV
BAC
C
AXP
AMZN
CSCO
CVX
COST
COP
CL
Technology
Healthcare
Finance
Consumer Staples
Energy
PC1
PC2


L1 Quantum Annealing
Energy: -0.5763
AMD
Technology
Healthcare
Finance
Consumer Staples
Energy
BMY
AAPL
AMZN
BAC
C
BLK
CVX
COST
CL
COP
PC1
PC2

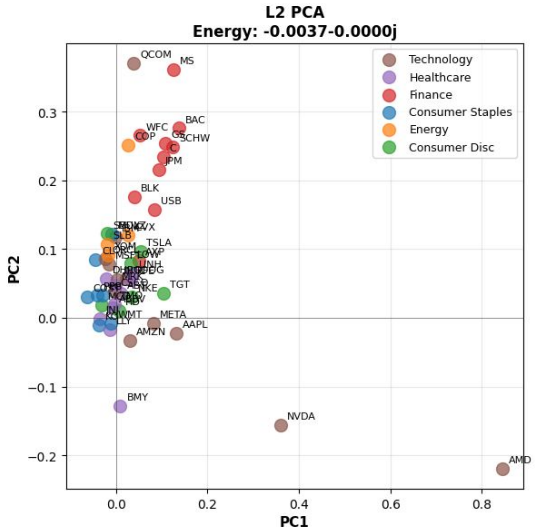

L1 Dual Annealing
Energy: -0.5763
AMD
Technology
Healthcare
Finance
Consumer Staples
Energy
BMY
AAPL
AMZN
BAC
C
BLK
CVX
COST
CL
COP
PC1
PC2

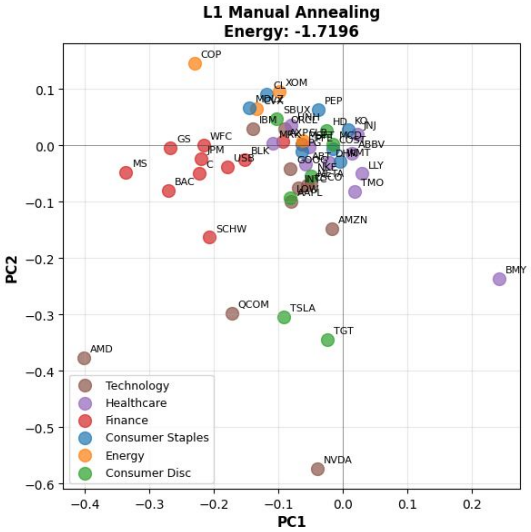

Multi-Component QAPCA
Energy: -0.0163
AMD
BAC
COP
C
AAPL
AXP
CSCO
ABT
AMZN
COST
ABBV
BMY
Technology
Healthcare
Finance
Consumer Staples
Energy
PC1
PC2

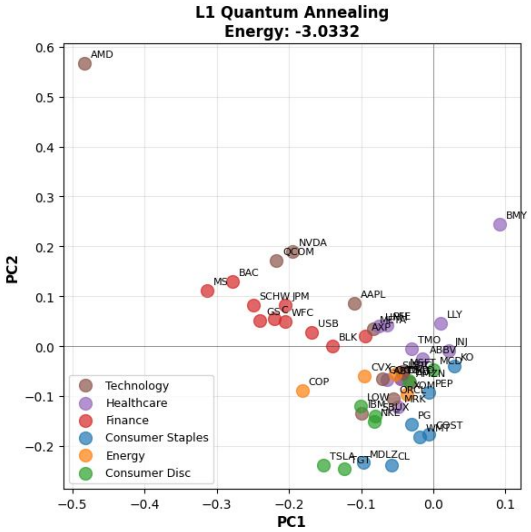
15 Tickers

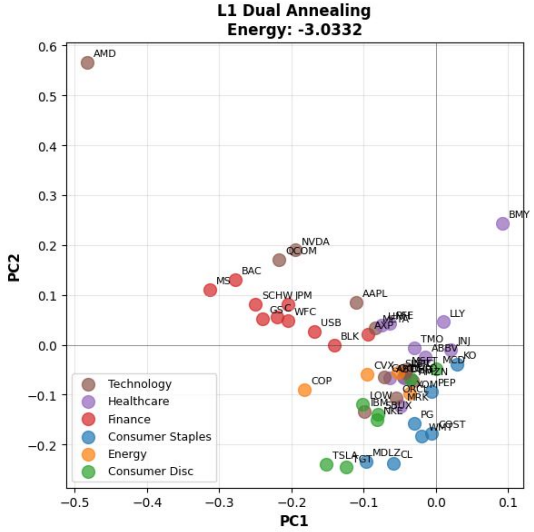

L2 PCA
Energy: -0.0028
L1 Manual Annealing
Energy: -0.9623
L1 Quantum Annealing
Energy: -1.2204
L1 Dual Annealing
Energy: -1.2204
Multi-Component QAPCA
Energy: -0.0377
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2

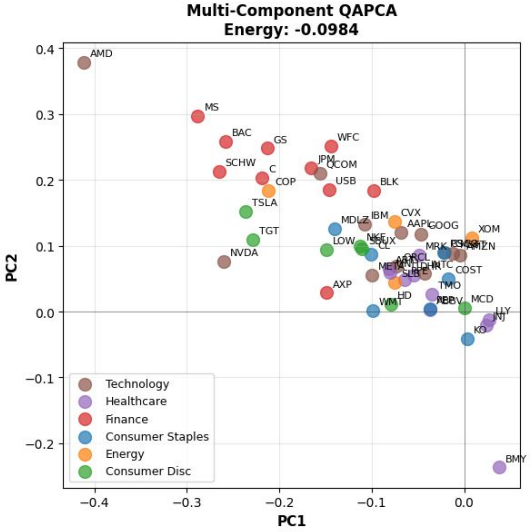
30 Tickers


L2 PCA
Energy: -0.0037-0.0000j
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2

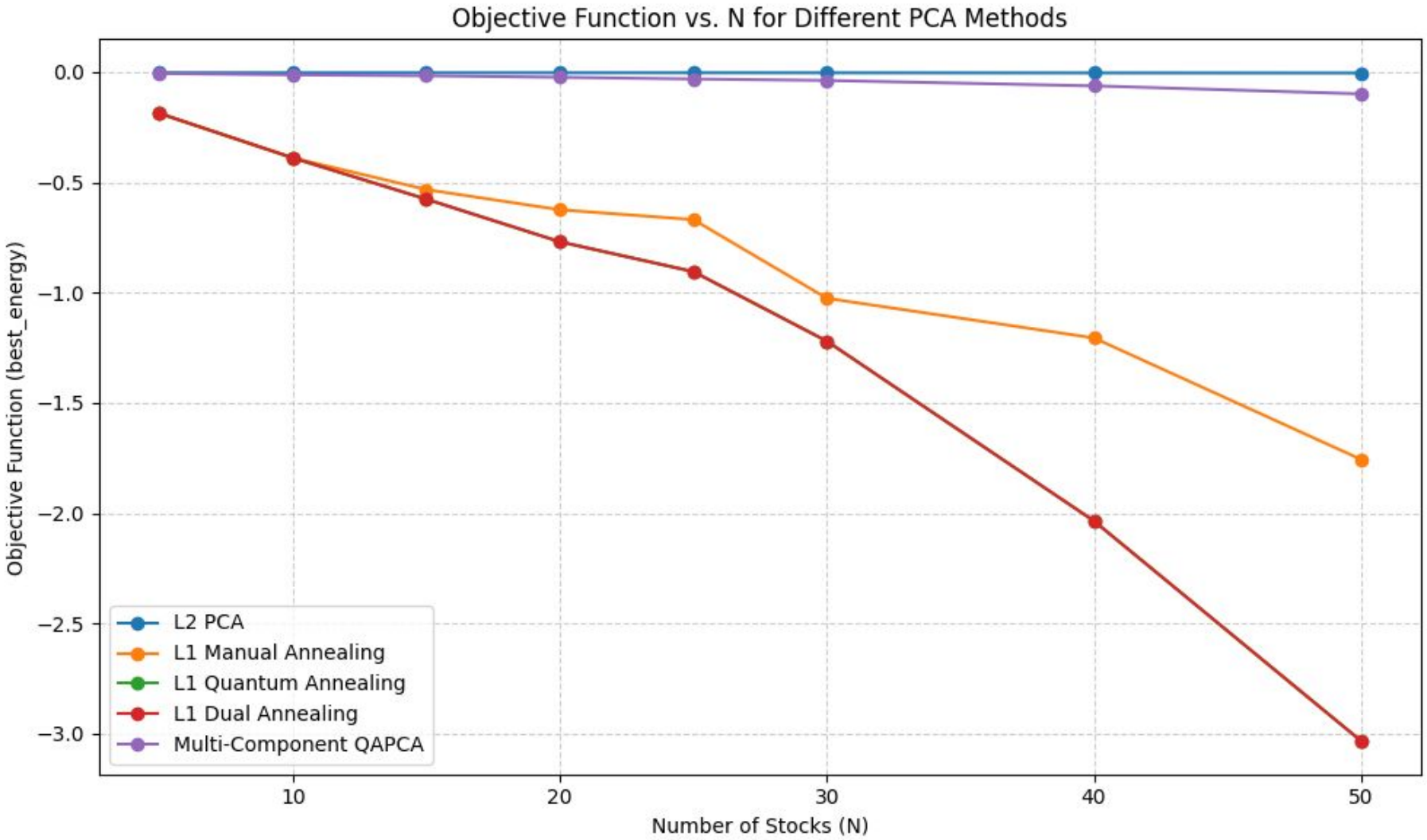

L1 Manual Annealing
Energy: -1.7196
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2

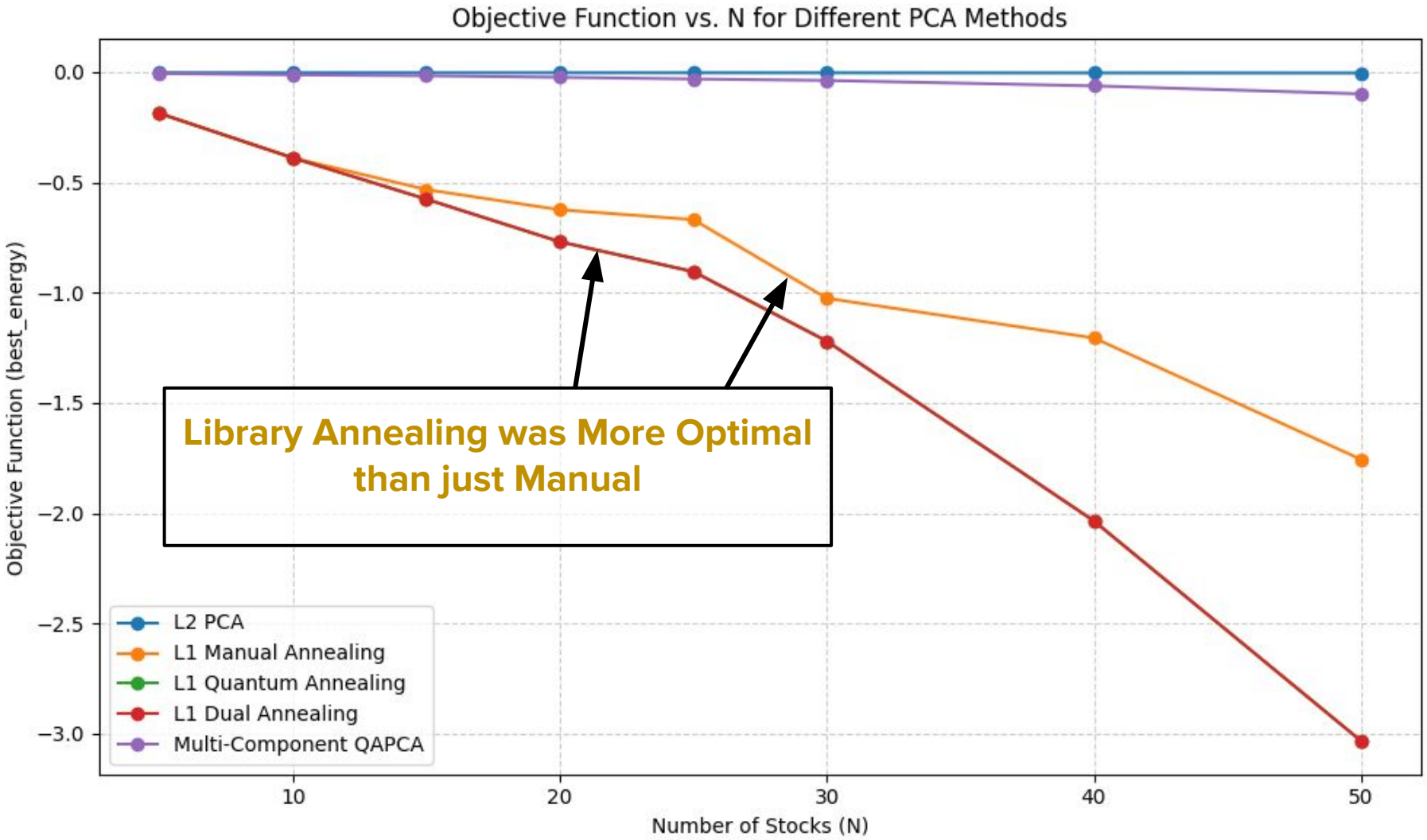

L1 Quantum Annealing
Energy: -3.0332
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2


L1 Dual Annealing
Energy: -3.0332
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2


Multi-Component QAPCA
Energy: -0.0984
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
PC1
PC2

50 Tickers


Objective Function vs. N for Different PCA Methods
Objective Function (best_energy)
Number of Stocks (N)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
10
20
30
40
50
L2 PCA
L1 Manual Annealing
L1 Quantum Annealing
L1 Dual Annealing
Multi-Component QAPCA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0.0
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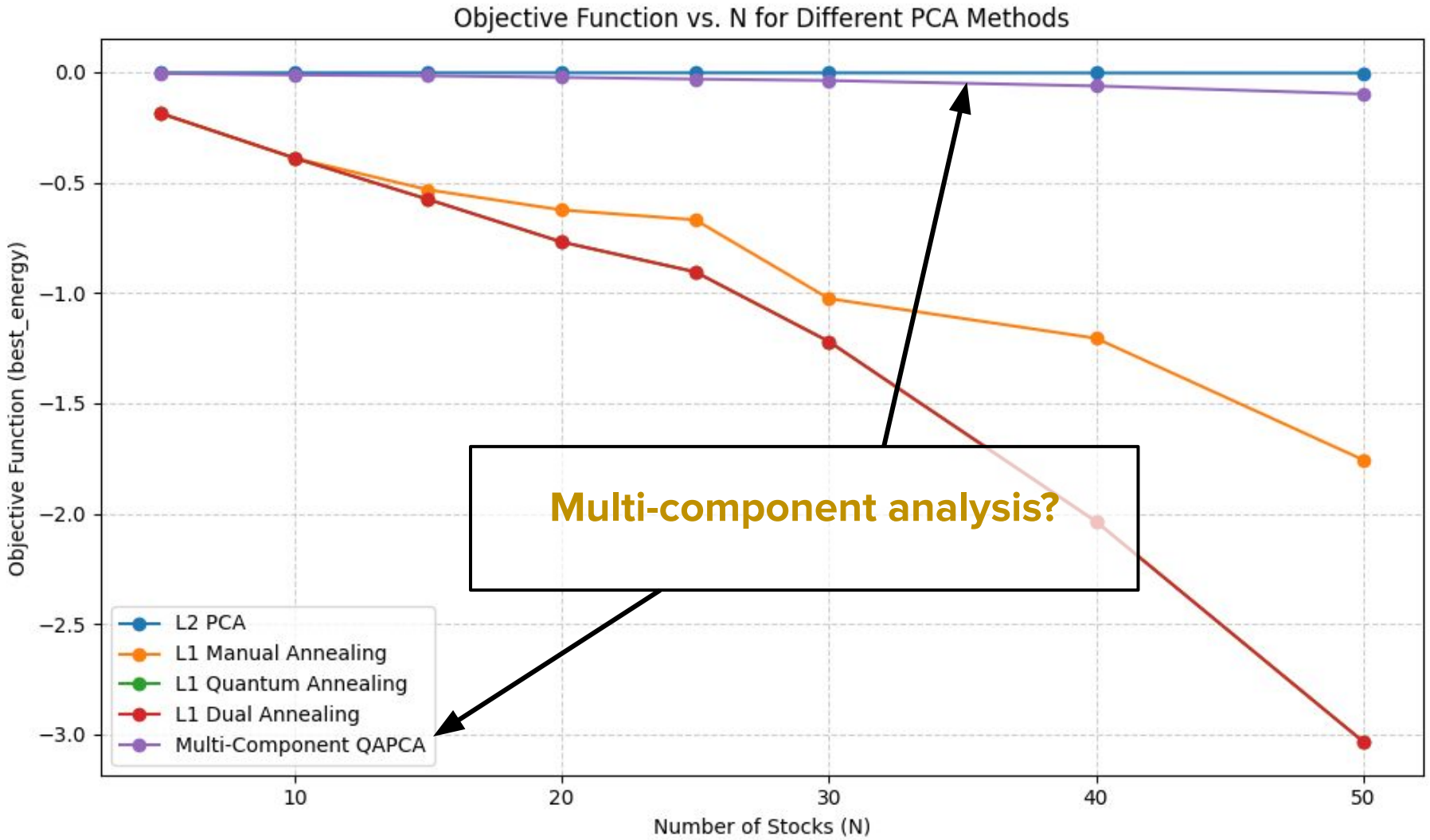
−2.5
−3.0
10
20
30
40
50
L2 PCA
L1 Manual Annealing
L1 Quantum Annealing
L1 Dual Annealing
Multi-Component QAPCA
Library Annealing was More Optimal than just Manual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0.0
−0.5
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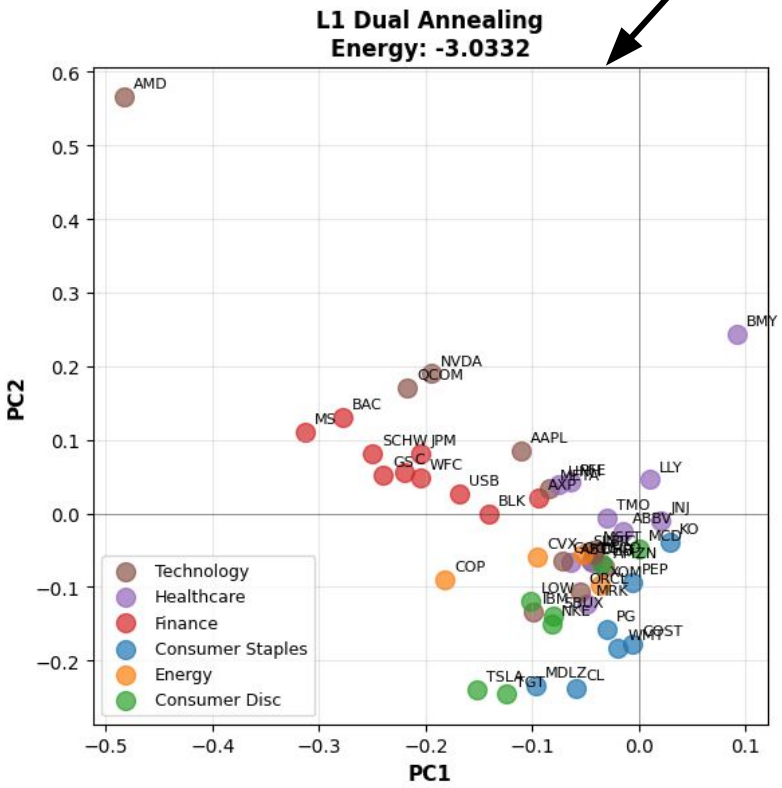
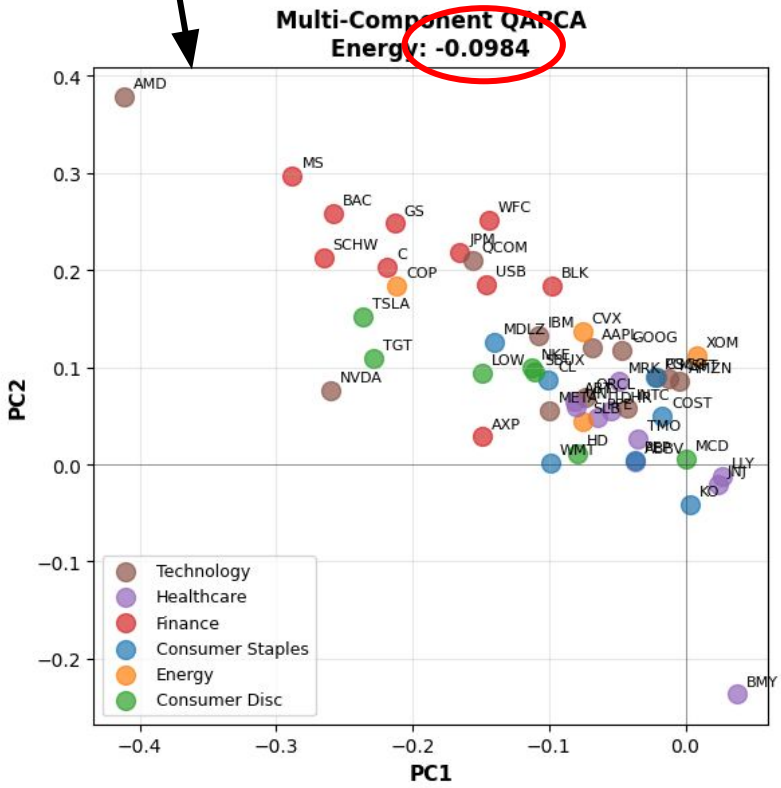
−3.0
10
20
30
40
50
L2 PCA
L1 Manual Annealing
L1 Quantum Annealing
L1 Dual Annealing
Multi-Component QAPCA
Multi-component analysis?

Multi-components

Ineffective Optimisation in Comparison





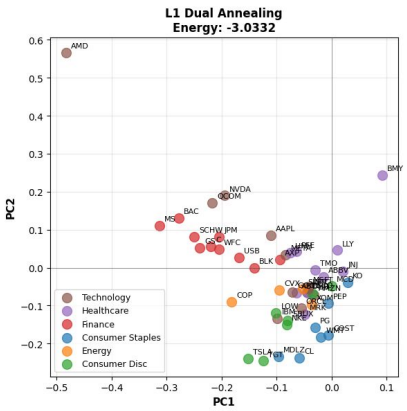
Multi-components

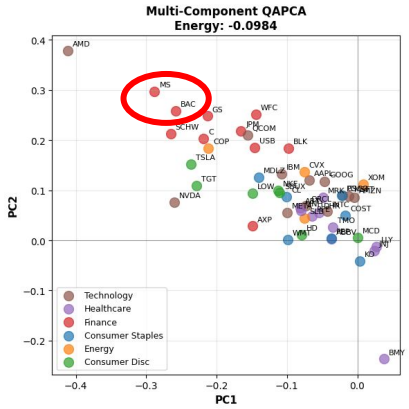
Possible reasons for worse optimization:

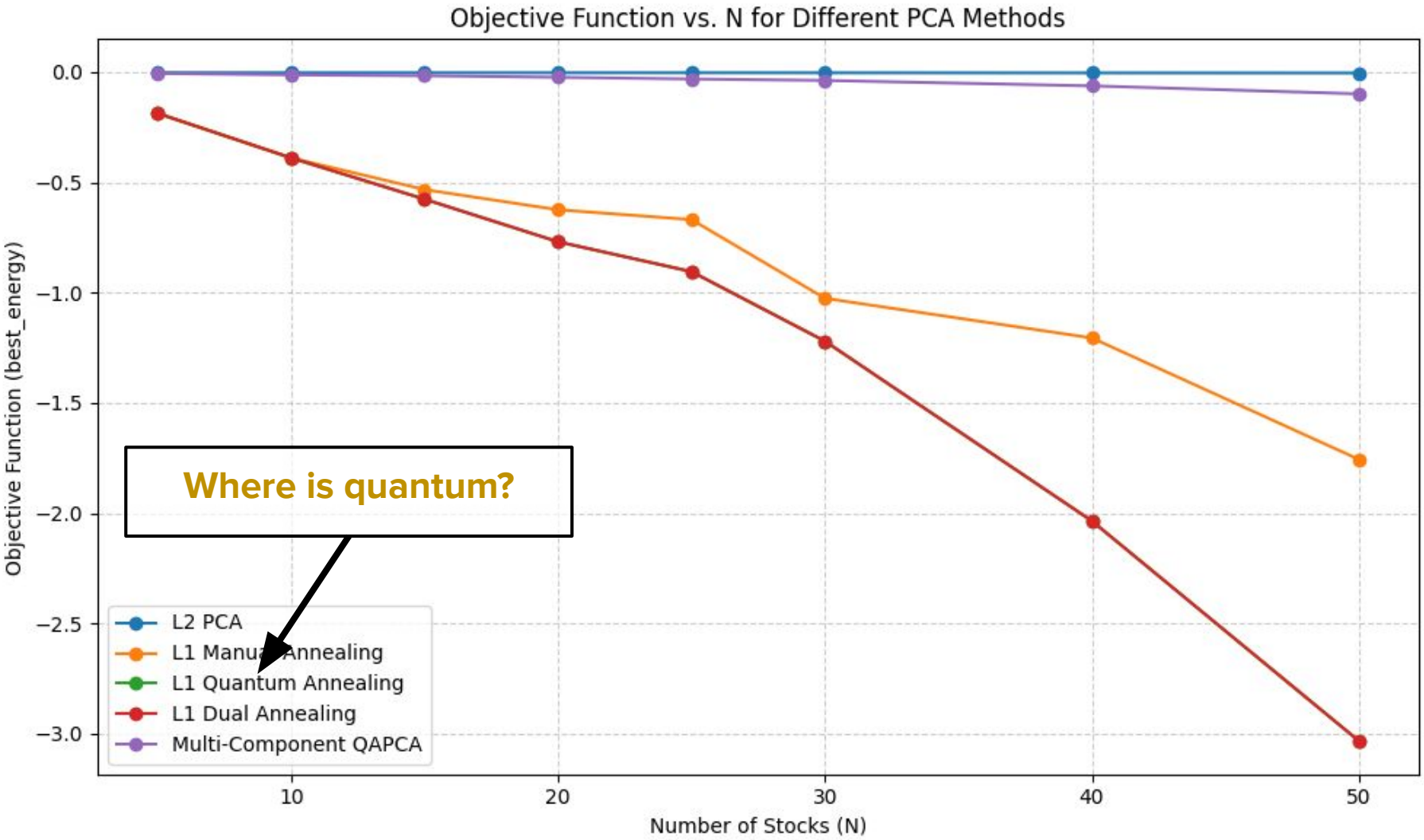
- **Greater possibility space requires more iterations.**
- **Altering epsilon oscillated between local minima.**

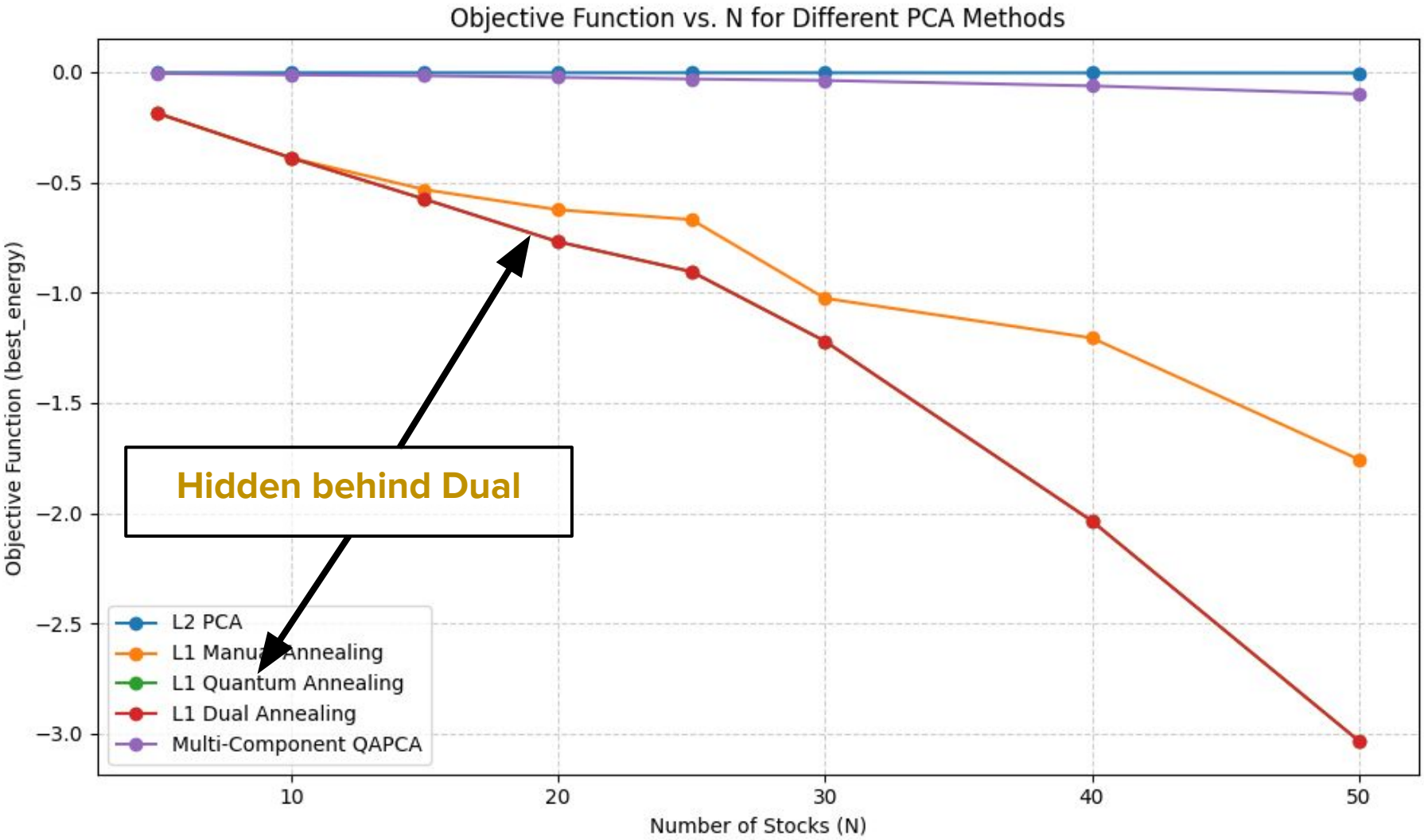
Conclusion:

- **Greater time complexity without improved accuracy.**
- **Mathematically interesting but not a practical improvement over other methods.**






Objective Function vs. N for Different PCA Methods
Objective Function (best_energy)
Number of Stocks (N)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
10
20
30
40
50
L2 PCA
L1 Manual Annealing
L1 Quantum Annealing
L1 Dual Annealing
Multi-Component QAPCA
Where is quantum?


Objective Function vs. N for Different PCA Methods
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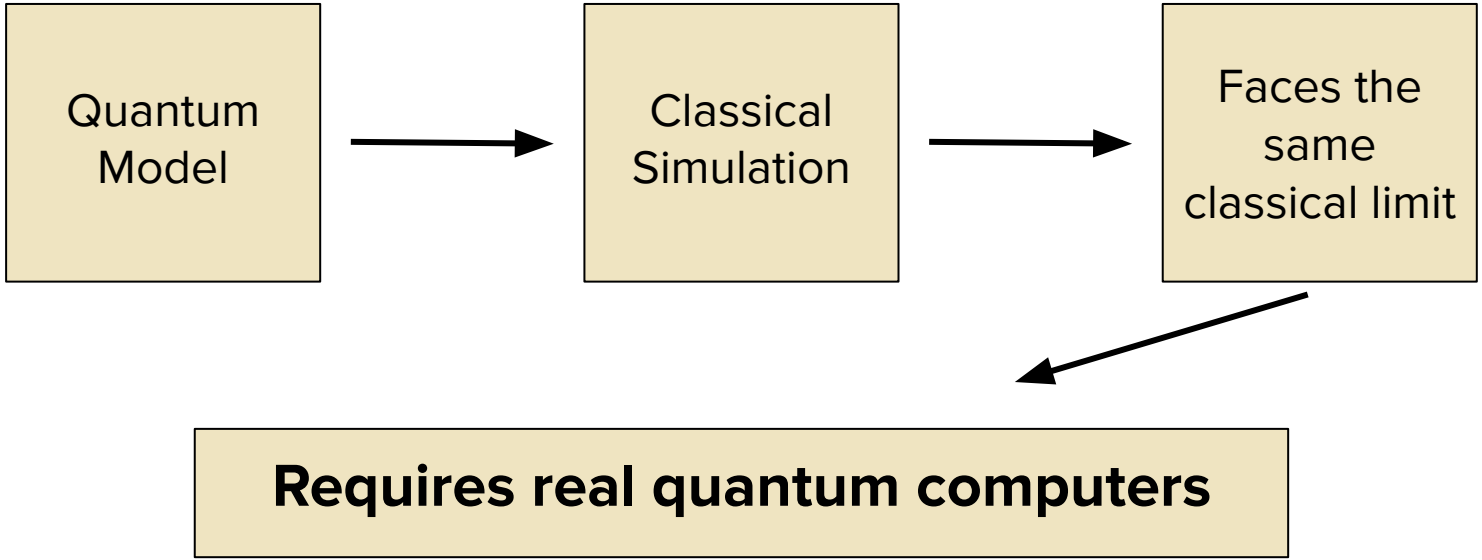
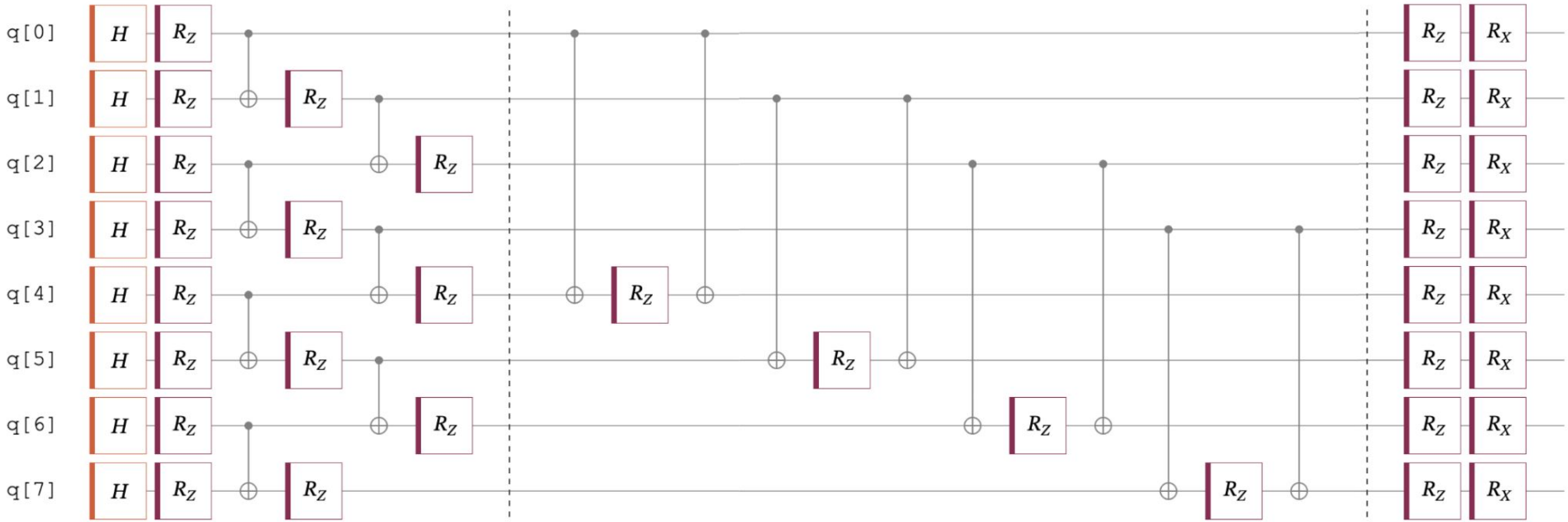
−3.0
10
20
30
40
50
L2 PCA
L1 Manual Annealing
L1 Quantum Annealing
L1 Dual Annealing
Multi-Component QAPCA
Hidden behind Dual

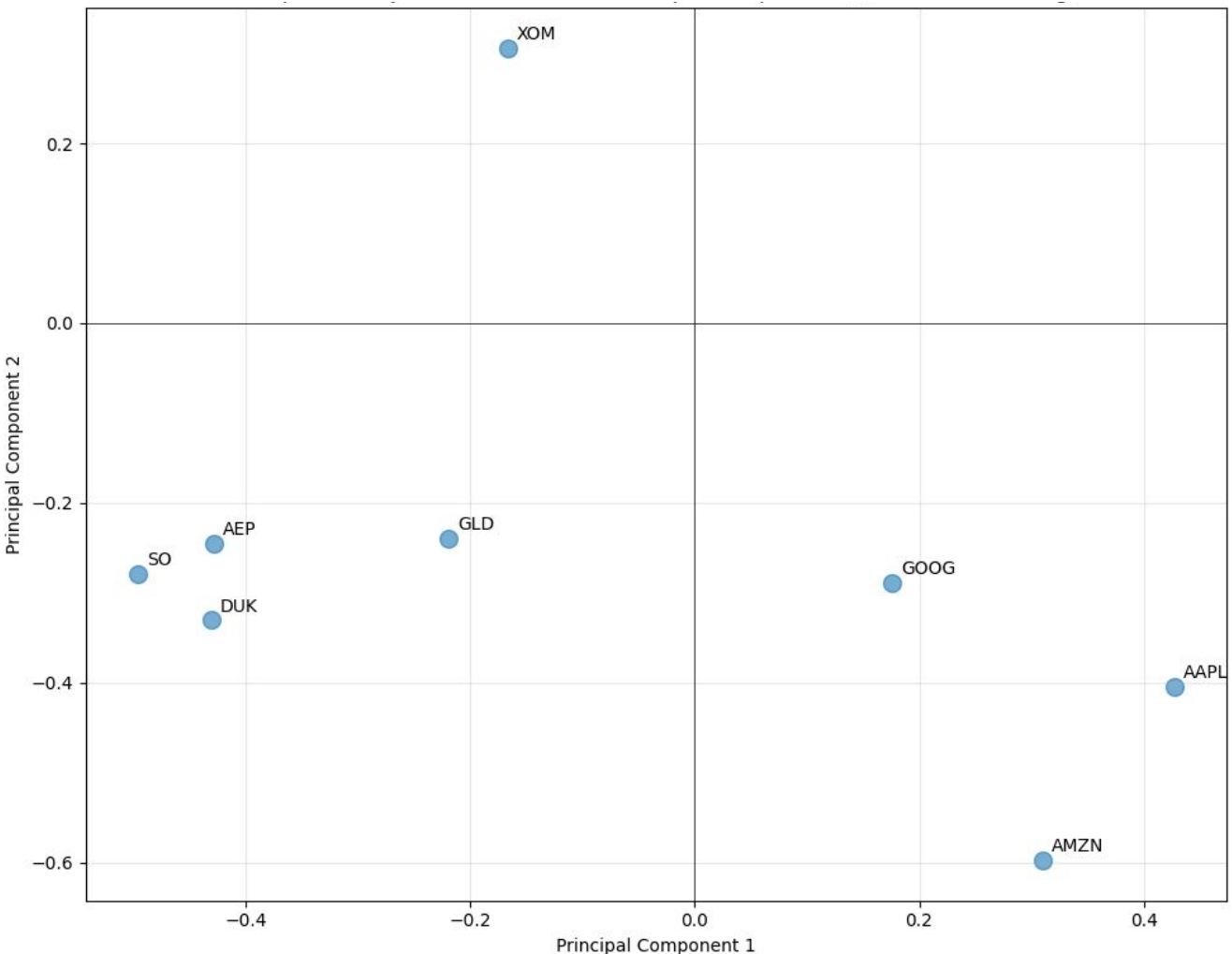
Why isn't Quantum working?



REAL quantum implementation - QAOA



Companies Projected onto First Two Principal Components (Qubits)



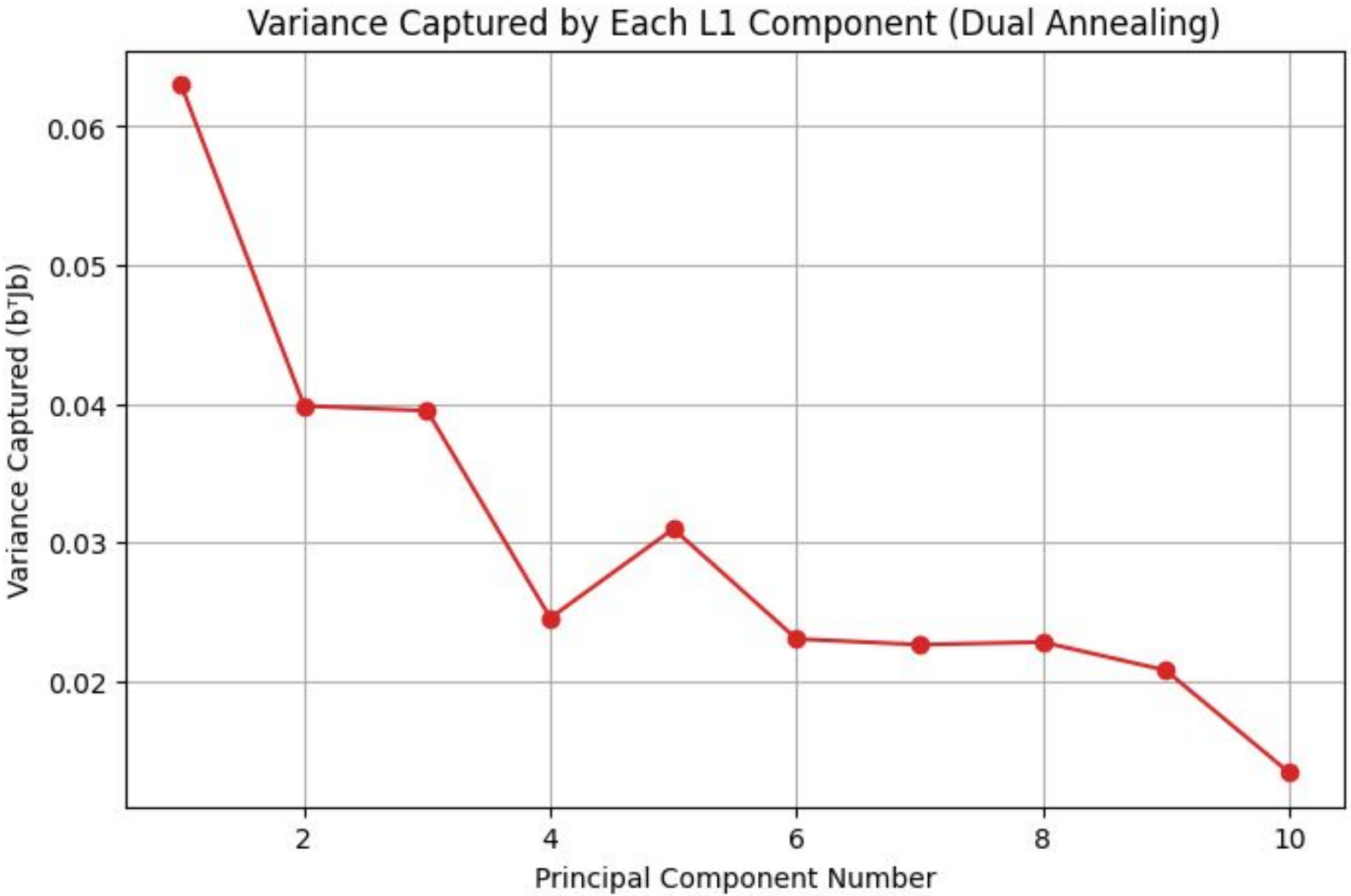
Relevance to financial firms and next steps

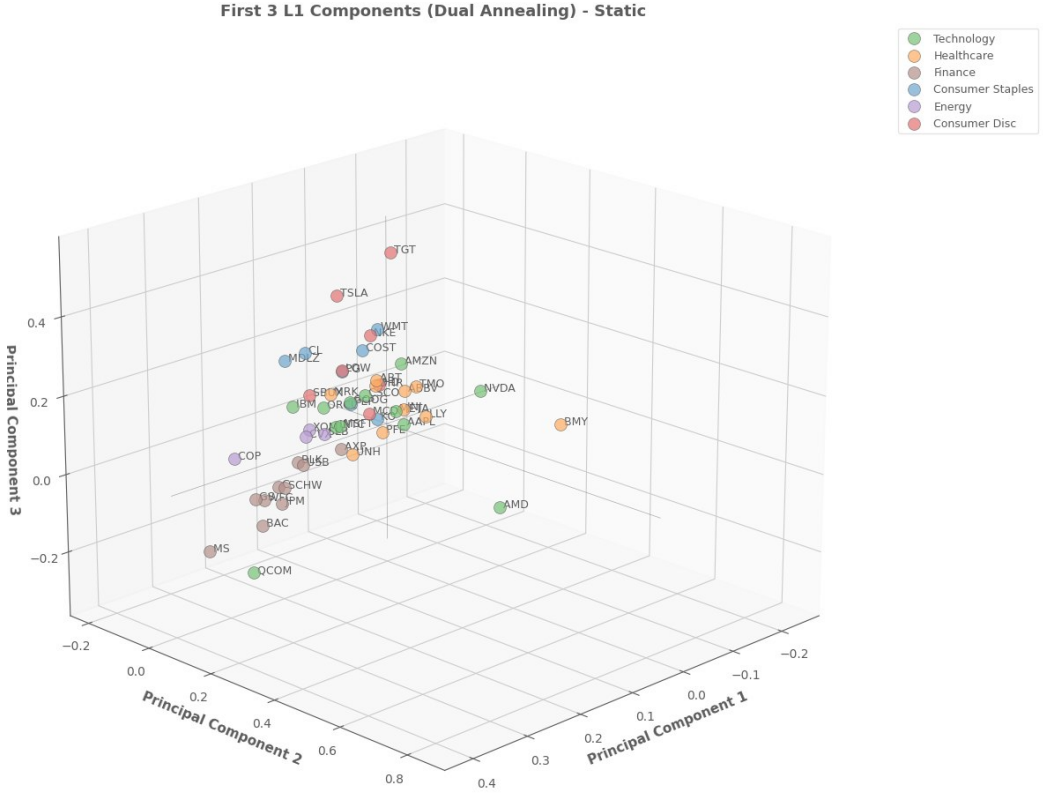
Principal component analysis benefits:

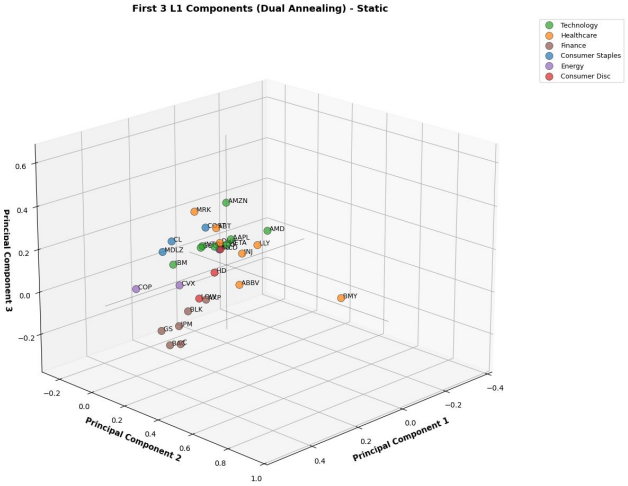
- Risk modeling
- Simplifies data (reduces noise)
- Helps optimize trades

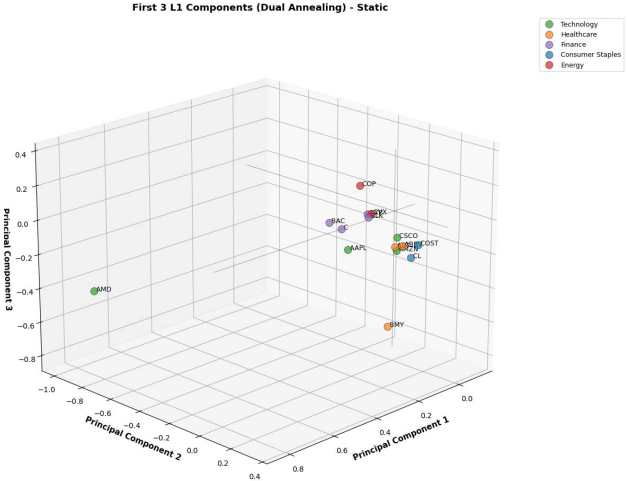
Next steps:

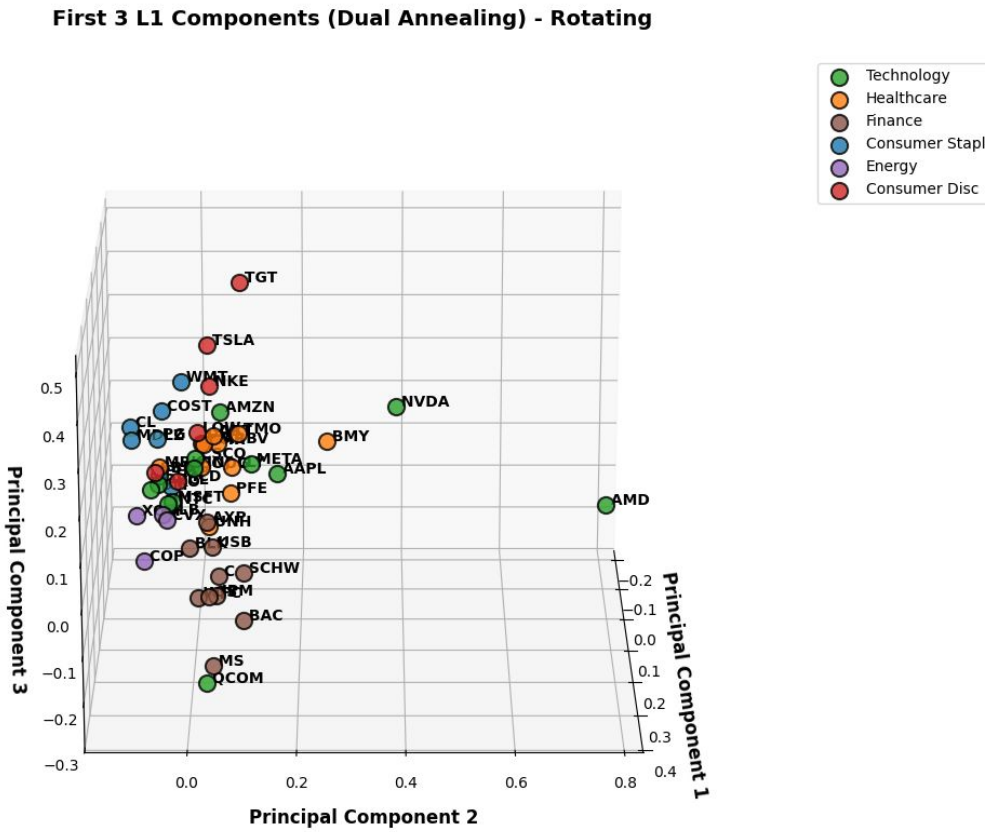
- Running on true quantum annealer
- Improving qubit count

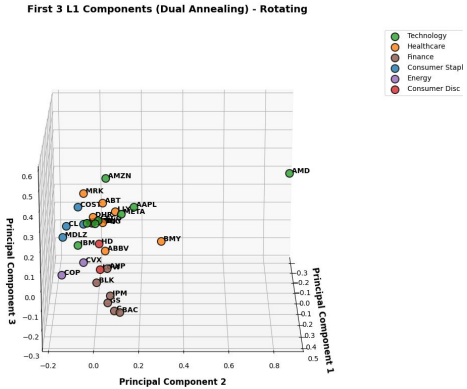

Variance Captured by Each L1 Component (Dual Annealing)
Variance Captured ($b^T Jb$)
0.06
0.05
0.04
0.03
0.02
2
4
6
8
10
Principal Component Number

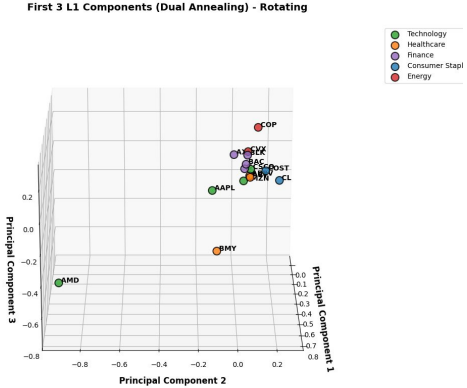

First 3 L1 Components (Dual Annealing) - Static
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
Principal Component 1
Principal Component 2
Principal Component 3
TGT
TSLA
WMT
COST
AMZN
NVDA
BMY
AMD
COP
MS
QCOM
BAC
JPM
SCHW
AXP
UNH
PFE
AAPL
LLY
MRK
IBM
SBUX
CL
MDLZ
PGW


First 3 L1 Components (Dual Annealing) - Static
Technology
Healthcare
Finance
Consumer Staples
Energy
Consumer Disc
Principal Component 1
Principal Component 2
Principal Component 3
AMZN
MRK
AMD
LLY
CL
MDLZ
IBM
HD
CVX
ABBV
COP
BMY
BLK
GS
PM
BAC


First 3 L1 Components (Dual Annealing) - Static
Technology
Healthcare
Finance
Consumer Staples
Energy
Principal Component 1
Principal Component 2
Principal Component 3
COP
BAC
CSCO
COST
AAPL
CL
AMD
BMY


First 3 L1 Components (Dual Annealing) - Rotating
Technology
Healthcare
Finance
Consumer Stapl
Energy
Consumer Disc
Principal Component 1
Principal Component 2
Principal Component 3


First 3 L1 Components (Dual Annealing) - Rotating
Technology
Healthcare
Finance
Consumer Stapl
Energy
Consumer Disc
Principal Component 1
Principal Component 2
Principal Component 3


First 3 L1 Components (Dual Annealing) - Rotating
Technology
Healthcare
Finance
Consumer Stapl
Energy
Principal Component 1
Principal Component 2
Principal Component 3

Quantum Music

Summary of Main Results:

- Implemented classical library-based annealer and manual annealer
- Tested with 50+ stocks, outliers, and longer time intervals
- Compared to classically simulated quantum annealing
- Rejected multicomponent annealing because of computational complexity without improved accuracy
- Using qiskit simulation, ran quantum annealing simulation on IBM QC

Plus:

- Implemented Sklearn and Lin-Alg eigenvector decomp for L2
- Found objective function vs N for each model
- Found variance captured by each component
- Found and plotted 3d Component
- Sonified data as Quantum music